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# %%
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# %%
# Q1: Ask user for value M and plot y = M*x^2 and y = M*sin(x)

M = float(input("Enter a value for M: "))

x = np.linspace(-10, 10, 100)
y1 = M * (x ** 2)
y2 = M * np.sin(x)

plt.figure(figsize=(8, 5))
plt.plot(x, y1, color="blue", linestyle="-", label="y = M*x^2")
plt.plot(x, y2, color="red", linestyle="--", label="y = M*sin(x)")
plt.title("Plot of y = M*x^2 and y = M*sin(x)")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.grid(True)
plt.show()

# %%
# Q2: Dataset of 5 subjects and scores using Seaborn bar plot

subjects = ["Math", "Physics", "Chemistry", "English", "Computer Science"]
scores = [85, 78, 92, 74, 88]

df_scores = pd.DataFrame({"Subject": subjects, "Score": scores})
print(df_scores)

plt.figure(figsize=(8, 5))
ax = sns.barplot(x="Subject", y="Score", data=df_scores, palette="viridis", hue="

# annotate each bar with its score
for i, score in enumerate(scores):
    ax.text(i, score + 1, str(score), ha="center")

plt.title("Scores of 5 Subjects")
plt.xlabel("Subject")
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plt.ylabel("Score")
plt.grid(True, axis="y")
plt.show()

# %%
# Q3: Use roll number as seed, generate 50 random values, 2x2 subplot layout

roll_number = 1024170089
np.random.seed(roll_number)

data = np.random.randn(50)
cumulative_sum = np.cumsum(data)
noise = np.random.randn(50)

fig, axs = plt.subplots(2, 2, figsize=(10, 8))

# Line plot: cumulative sum
axs[0, 0].plot(cumulative_sum, color="green")
axs[0, 0].set_title("Cumulative Sum")
axs[0, 0].set_xlabel("Index")
axs[0, 0].set_ylabel("Cumulative Value")
axs[0, 0].grid(True)

# Scatter plot: random noise
axs[0, 1].scatter(range(50), noise, color="purple")
axs[0, 1].set_title("Random Noise (Scatter)")
axs[0, 1].set_xlabel("Index")
axs[0, 1].set_ylabel("Value")
axs[0, 1].grid(True)

# Extra plot: original random data as a line
axs[1, 0].plot(data, color="orange")
axs[1, 0].set_title("Original Random Data")
axs[1, 0].set_xlabel("Index")
axs[1, 0].set_ylabel("Value")
axs[1, 0].grid(True)

# Extra plot: histogram of data
axs[1, 1].hist(data, bins=10, color="teal")
axs[1, 1].set_title("Histogram of Random Data")
axs[1, 1].set_xlabel("Value")
axs[1, 1].set_ylabel("Frequency")
axs[1, 1].grid(True)

plt.tight_layout()
plt.show()
```

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# %%
# Q4: Company sales data analysis using Seaborn

url = "https://raw.githubusercontent.com/AnjulaMehto/MCA/main/company_sales_data.
df = pd.read_csv(url)
print(df.head())
print(df.columns)

# 1. Total profit of all months - line plot
plt.figure(figsize=(8, 5))
sns.lineplot(x="month_number", y="total_profit", data=df, marker="o", color="blue
plt.title("Total Profit of All Months")
plt.xlabel("Month Number")
plt.ylabel("Total Profit")
plt.grid(True)
plt.show()

# 2. All product sales data - multiline plot
products = ["facecream", "facewash", "toothpaste", "bathingsoap", "shampoo", "moi

plt.figure(figsize=(10, 6))
for product in products:
    sns.lineplot(x="month_number", y=product, data=df, marker="o", label=product)

plt.title("Product Sales Data (All Months)")
plt.xlabel("Month Number")
plt.ylabel("Units Sold")
plt.legend()
plt.grid(True)
plt.show()

# 3. Bar chart for all features/attributes (average value of each column)
plt.figure(figsize=(10, 6))
avg_values = df.drop(columns=["month_number"]).mean()
sns.barplot(x=avg_values.index, y=avg_values.values, palette="Set2", hue=avg_valu
plt.title("Average Value of Each Feature/Attribute")
plt.xlabel("Feature")
plt.ylabel("Average Value")
plt.xticks(rotation=45)
plt.grid(True, axis="y")
plt.tight_layout()
plt.show()
```